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SH.P.K

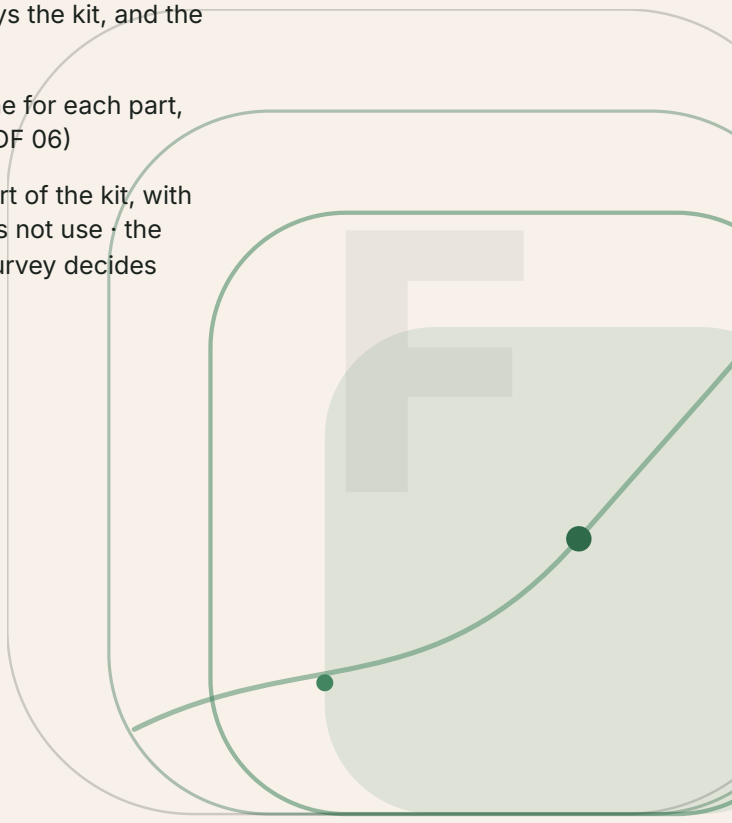
F

HARDWARE LIST · KIT F

Hardware for a farm or greenhouse with irrigation

Every part of the kit for a farm or a greenhouse with irrigation: what each part does there, how many, how it qualifies, and which sheet to open.

KIND OF SITE	Farm or greenhouse with irrigation: soil and weather, with the water supply behind it
FOR	The project manager who plans and buys the kit, and the founder who approves the purchase
GOES WITH	The hardware sheets HW01 to HW17, one for each part, and the site pack for this kind of site (PDF 06)
COVERS	What is measured and where · every part of the kit, with its sheet · the parts this kind of site does not use · the numbers on the data cable · what the survey decides



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KIT F

Hardware for a farm or greenhouse with irrigation

A farm or a greenhouse with irrigation: the soil and the air at the field, with the water supply behind it. Greecon measures the moisture of the soil in each irrigation zone, the air temperature and humidity, how full the tank is, the pressure in the irrigation line and, if there is a pump to watch, the pump's flow and whether it runs. This list is every part of the kit for that, with the sheet that says how each one qualifies.

What is measured, and where

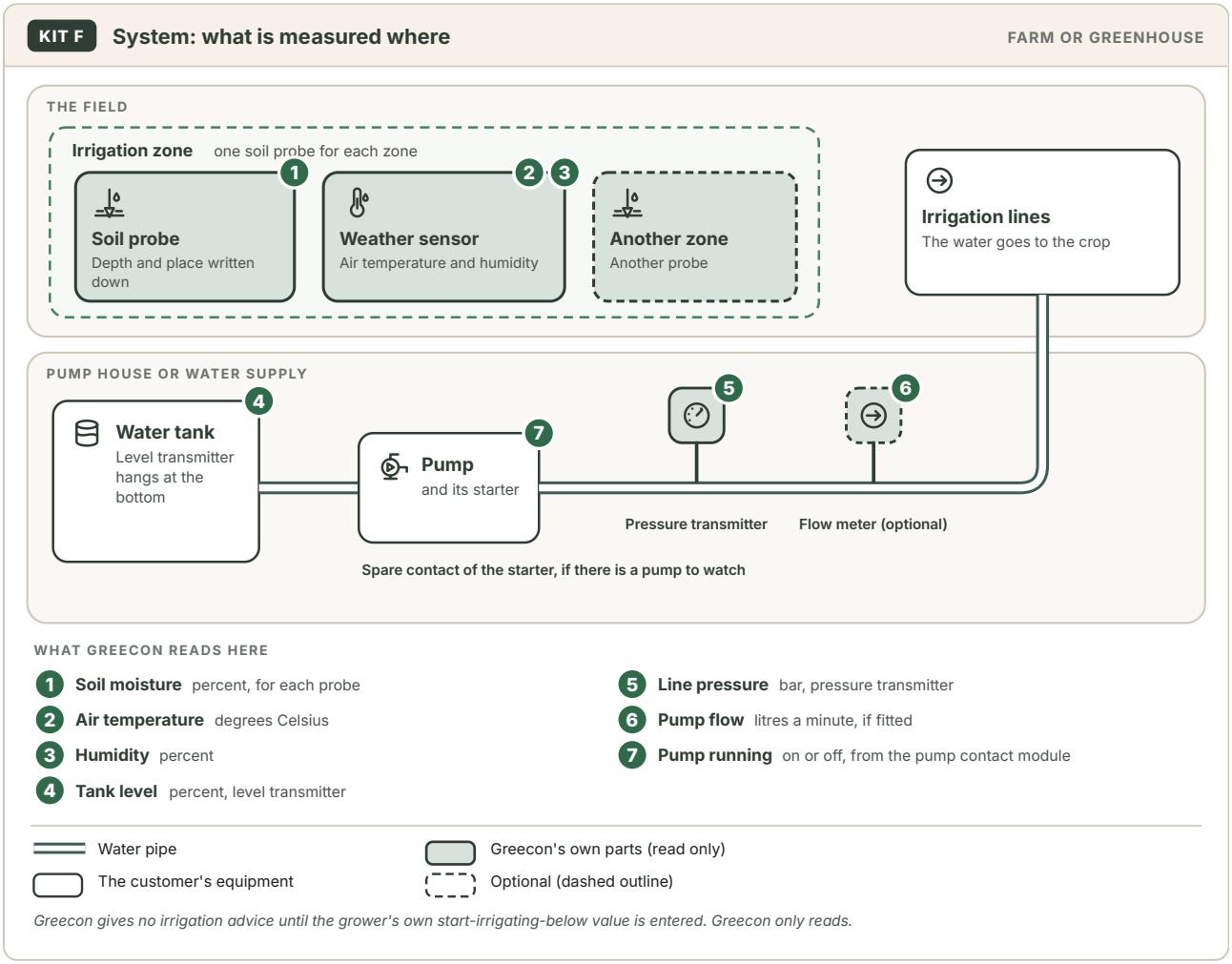


Figure F-1. Kit F, farm or greenhouse with irrigation. System: what is measured where. The dashed parts are optional.

The kit, part by part

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW01	**Box** By specification · EUR 180 to 300	IN THE KIT	1	In the pump house. It reads the soil probes, the weather sensor, the water module and, if there is a pump to watch, the pump contact module. A 4G router is the likelier way to the internet.
HW02	**Converter** By specification · EUR 30 to 60 (basic) or 250 and up (industrial)	IN THE KIT	1	In the enclosure in the pump house. Behind it: the water module (3), the pump contact module (5) if there is a pump to watch, the soil probe (6; a second probe takes 8) and the weather sensor (7).
HW03	**Data cable and labels** By specification · EUR 40 to 80	IN THE KIT	1 lot	The longest run of the four: from the converter in the pump house through the soil probe and the weather sensor in the field and the water module to the pump contact module. Plan it under 100 metres.
HW04	**Cellular router** By specification · EUR 150 to 200	IN THE KIT	0 or 1	The likelier case: a field or a pump house seldom has a cable to the internet. The box and the converter plug into the router's sockets.
HW08	**Waveshare 4 to 20 mA module** Approved model · EUR 30 to 60	IN THE KIT	1	One, for the tank and the pipe behind the irrigation. The flow meter (input 3) only if there is a pump to watch.
HW09	**Waveshare pump contact module** Approved model · EUR 30 to 60	OPTIONAL	0 or 1	Only if there is a pump to watch and its starter has a spare running contact.
HW10	**Level transmitter** By specification · EUR 120 to 400	IN THE KIT	1	One, in the irrigation tank or reservoir. It gives Tank level, with the standard low-level alerts.
HW11	**Pressure transmitter** By specification · EUR 40 to 150	IN THE KIT	1	One, on the irrigation line.
HW12	**Flow meter** By specification · EUR 150 to 500	OPTIONAL	0 or 1	Only if there is a pump to watch.
HW13	**Soil moisture probe** By register table · EUR 40 to 120 each	IN THE KIT	1 for each zone	One for each irrigation zone, at the depth and in the place agreed with the grower. It gives Soil moisture. Greecon gives no irrigation advice until the grower's own Start irrigating below value is entered.

Sheet	Hardware	In the kit	How many	What it does at this kind of site
HW14	**Temperature and humidity sensor** By register table · EUR 20 to 80	IN THE KIT	1	One, at the field or in the greenhouse. It gives Air temperature and Humidity. The frost alert says nothing until the grower's own Frost alert at or below value is entered.
HW15	**24 V power supply** By specification · EUR 25 to 50	IN THE KIT	1	One, as at a hotel: the modules, the loops and the converter.
HW17	**Enclosure and fittings** By specification · EUR 80 to 150	IN THE KIT	1	One, in the pump house: as at a hotel, with the 4G router on the rail. The probes and the weather sensor are in the field, on the data cable.

About EUR 700 to 1,500. The figures are planning estimates, not quotes. One soil probe is bought for each irrigation zone.

Sheet	Hardware	Why it is not in this kit
HW05	**Eastron SDM630 three-phase meter**	A farm kit has no energy meter. One can be added to any site from Add equipment Greecon already knows, and then this sheet applies.
HW06	**Eastron SDM230 single-phase meter**	A farm kit has no energy meter.
HW16	**UPS**	Not in the standard list for this kit. Decide at the survey whether this site gets one, and write the decision down.

Never at this kind of site: HW07 Battery system.

The numbers on the data cable

Number	Device	What it is	Sheet
3	4 to 20 mA module	Waveshare Modbus RTU Analog Input 8CH. The platform calls it the Water I/O module	HW08
5	Pump contact module	Waveshare Modbus RTU IO 8CH	HW09
6	Soil probe	Any probe that qualifies by its register table. A second probe takes the next free number, 8	HW13
7	Weather sensor	Any temperature and humidity sensor that qualifies by its register table	HW14

A second soil probe, for a second zone, takes the next free number: 8, then 9. The numbers of the probes and of the weather sensor are set **before the probe is buried and the sensor is mounted.** Everything on the cable is set to 9600 baud, 8 data bits, no parity, 1 stop bit, and the converter to port 502, in Modbus TCP to RTU gateway mode.

What the survey decides

Ask, or look	It decides
How many irrigation zones are there?	How many soil probes (sheet HW13)
Which crop, which soil, how deep are the roots, and which spot is typical of the zone?	Where each probe goes and how deep. The irrigation limit is the grower's own, and is asked after the first week
At what air temperature does frost damage the crop, and in which weeks?	The frost limit, and who is woken (sheet HW14)
How tall is the tank inside, and what pressure does the line see?	The ranges of the level and pressure transmitters (sheets HW10 and HW11)
Is there a pump to watch, and has its starter a spare contact that follows it?	The pump contact module and the flow meter (sheets HW09 and HW12)
Is there a phone signal at the pump house, or a network cable to it?	The cellular router, which is the likelier case in a field (sheet HW04)
How far is it from the pump house to the farthest probe?	The data cable: plan the whole run under 100 metres (sheet HW03)
Does this site get a UPS?	The standard list for this kit has none. Decide, and write the decision down (sheet HW16)

The whole job at this kind of site, from the survey to the handover, is in the site pack for a farm or greenhouse (PDF 06).